

COBALT

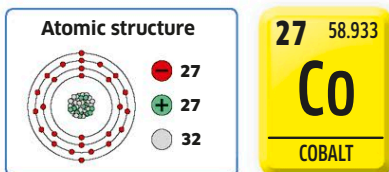
Cobaltum

Discovered: 1739

Cobalt is somewhat similar to iron, its neighbor on the periodic table. The metal is often added to alloys, including those used to make permanent magnets. A cobalt compound has long been used to produce "cobalt blue," a deep, vibrant blue for paints and dyes.

Cobalt color

Extracted from its ore, pure cobalt metal is silvery gray in appearance.



CADMIUM

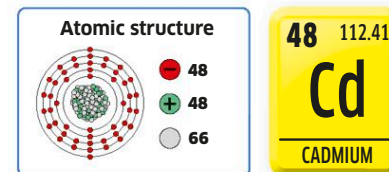
Cadmium

Discovered: 1817

Although it has some uses in industry and laser technology, this metal is now known to be highly toxic and dangerous to humans. If ingested, it can react like calcium, an essential and useful element, but will replace the calcium in our bones. This causes bones to become soft and easy to break.

Poisonous pellet

This sample of pure cadmium has been refined in a laboratory.



NICKEL

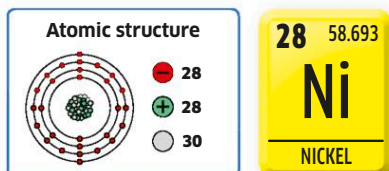
Niccolum

Discovered: 1751

This useful metal, which does not rust, is one of the ingredients in stainless steel (see p.63). It is also used to protect ships' propellers from rusting in water. Its best-known role is perhaps in the various alloys used to make coins, including the US 5-cent coin that is called a nickel.

Pure nickel

These samples of pure nickel have been shaped into tiny balls.



MERCURY

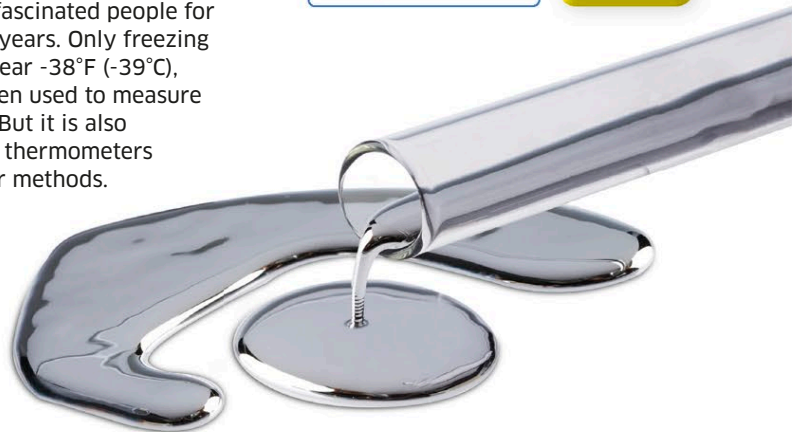
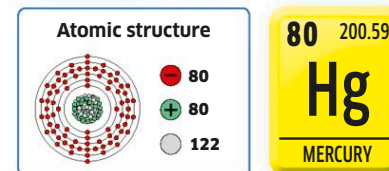
Hydrargyrum

Discovered: 1500 BCE

Famous for being the only metal that is liquid at room temperature, mercury has fascinated people for thousands of years. Only freezing to a solid at near -38°F (-39°C), it has long been used to measure temperature. But it is also poisonous, so thermometers now use other methods.

Quick liquid

Mercury is also known as quicksilver, and it is easy to see why.



TITANIUM

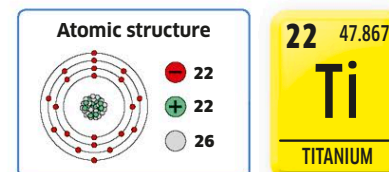
Titanium

Discovered: 1791

Known for its strength, this metal was named after the Titans, the divine and tremendously forceful giants of Greek mythology. Titanium is hard but also lightweight, and resistant to corrosion. This super combination of properties makes it perfect for use in artificial joints and surgical pins, but also in watches and in alloys for the aerospace industry. It is, however, a very expensive material.

Laboratory sample

Although titanium is a common element in Earth's crust, it usually only exists in mineral compounds, not as a native element. Pure titanium has to be extracted and refined.



When titanium reacts with oxygen in the air it gets a duller gray coating. This actually works as a protection against corrosion.

